VIPP Report

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Introduction

In this report, I evaluated the `MRI-Prostate-Gland-and-Zone-Segmentor` model for prostate MRI segmentation. The model repository can be found at this GitHub page. I used the `PROMISE12_nifti` dataset and evaluated the first 25 cases, from `Case00` to `Case24`. Each case contains a prostate MRI scan and its corresponding segmentation mask.

Since the PROMISE12 ground truth used in this experiment consists of **whole-gland prostate annotations**, the evaluation was carried out as a **whole-gland segmentation task**. For each case, the predicted whole-gland mask was compared with the corresponding ground-truth mask using the Dice score. At the end of the evaluation, I obtained an average **whole-gland Dice score of 0.8599** over the 25 evaluated cases.

Qualitative Comparison in 3D Slicer

The following screenshots were taken from 3D Slicer to visually compare the provided ground-truth annotation and the model prediction. In these screenshots, the **annotation is shown in green**, while the **model output is shown in red**.

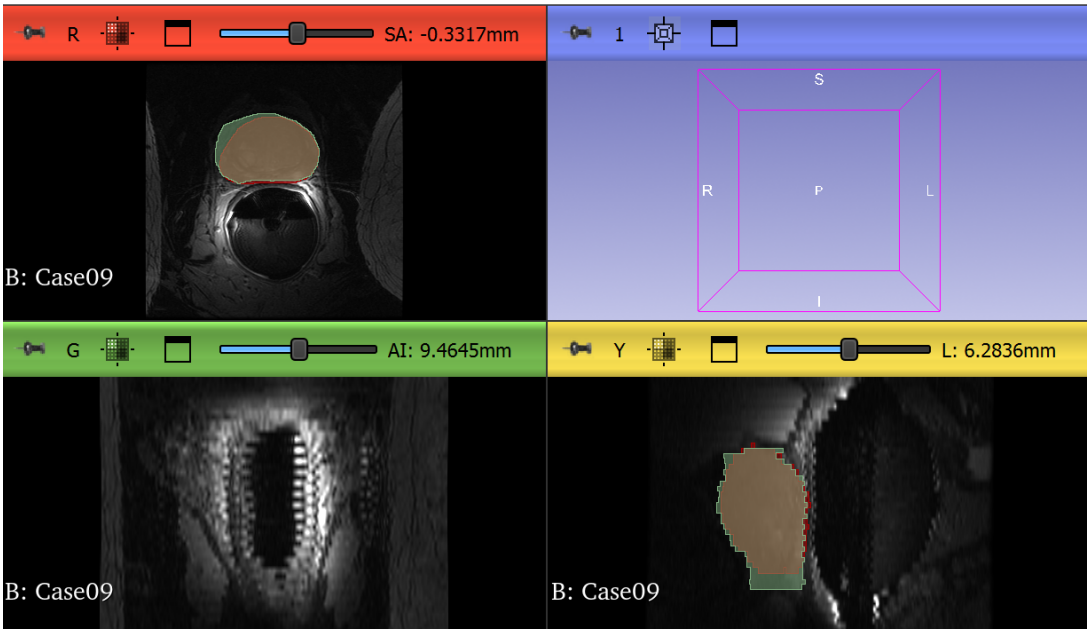

Figure 1: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

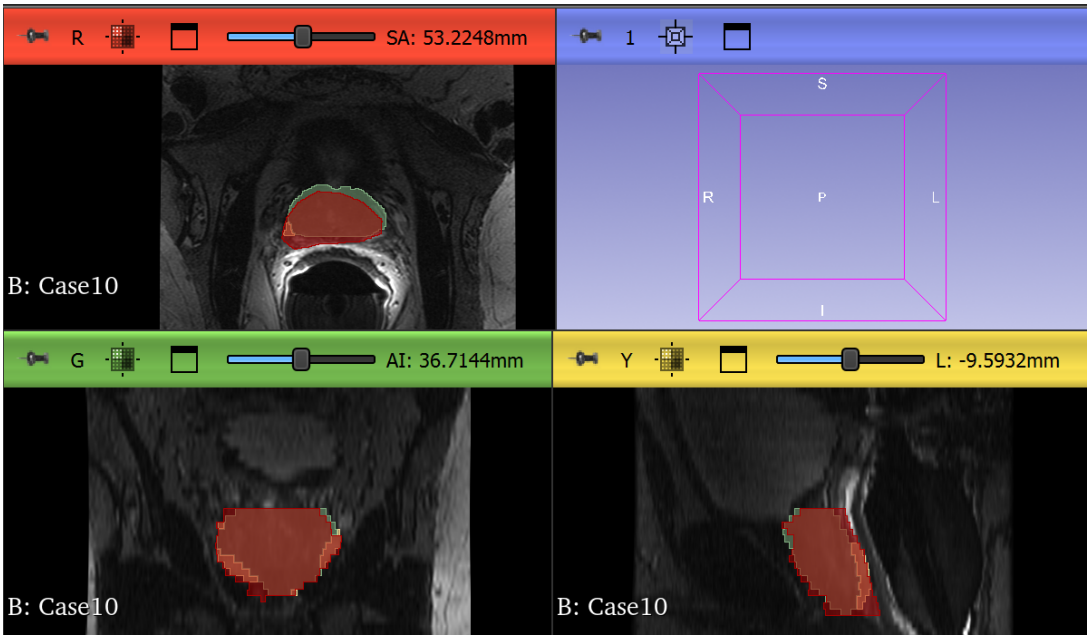

Figure 2: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

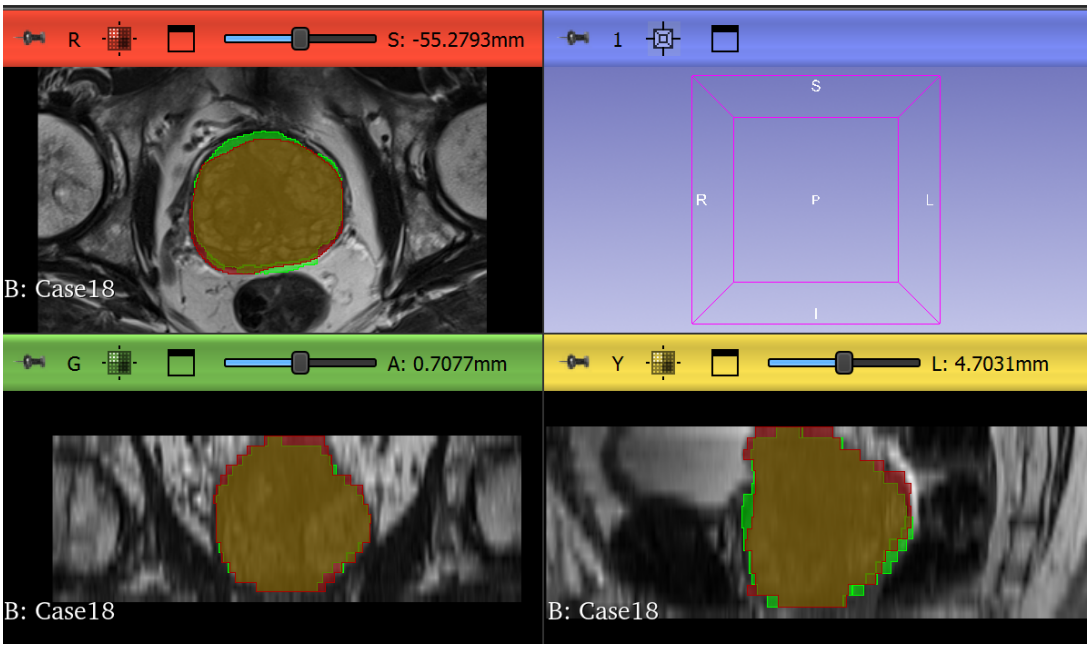

Figure 3: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

Quantitative Results

Case	Mode	Whole Gland Dice
Case00	whole-gland binary GT	0.873866
Case01	whole-gland binary GT	0.882352
Case03	whole-gland binary GT	0.912359
Case04	whole-gland binary GT	0.804384
Case05	whole-gland binary GT	0.859177
Case06	whole-gland binary GT	0.856169
Case07	whole-gland binary GT	0.869291
Case08	whole-gland binary GT	0.841655
Case09	whole-gland binary GT	0.883679
Case10	whole-gland binary GT	0.867200
Case11	whole-gland binary GT	0.910452
Case12	whole-gland binary GT	0.872310
Case13	whole-gland binary GT	0.890214
Case14	whole-gland binary GT	0.939916
Case15	whole-gland binary GT	0.925835
Case16	whole-gland binary GT	0.910074
Case17	whole-gland binary GT	0.877532
Case18	whole-gland binary GT	0.951042
Case19	whole-gland binary GT	0.927242
Case20	whole-gland binary GT	0.877388
Case21	whole-gland binary GT	0.919966
Case22	whole-gland binary GT	0.932685
Case23	whole-gland binary GT	0.957833
Case24	whole-gland binary GT	0.955024
Dataset mean	whole-gland binary GT	0.895700

Table 1: Dice score results obtained on the first 25 cases of the `PROMISE12_nifti` dataset. Since the ground truth consists of binary whole-gland annotations, zonal Dice scores are not applicable.

Discussion

The obtained results show that the model achieved generally strong whole-gland segmentation performance on most of the evaluated PROMISE12 cases. Most Dice scores are

high, with several cases above 0.90. This suggests that the pipeline worked well overall.